

Mathematics A

Unit 1 • Chapter OL-T50 Classified

Cross-session • chapter-organised candidate

8 classified items • 35 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

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Contents

Unit 1 • Unit 1 • Chapter OL-T50 • 8 items • 35 marks

Chapter	Items	Marks	Starts
01. OL-T50 3D Pythagoras & Trigonometry	8	35	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

3D Pythagoras & Trigonometry

Unit 1 • 35 marks • 8 item(s)

June 2025 | 4MA1-2H Q24 | 4 marks

Use midpoint or symmetry structure in a 3D model

January 2022 | 4MA1-1H Q21 | 5 marks

Use midpoint or symmetry structure in a 3D model

January 2020 | 4MA1-1H Q19 | 4 marks

Find angle between a line and a plane

January 2021 | 4MA1-2H Q17 | 3 marks

Find angle between a line and a plane

June 2021 | 4MA1-1H Q22 | 4 marks

Use a non-rectangular base before a line-plane angle

May-Nov 2020 | 2H Q18 | 5 marks

Find angle between a line and a plane

January 2023 | 2H Q23 | 5 marks

Use 3D trigonometry to obtain a terminal volume

November 2025 | 1H Q26 | 5 marks

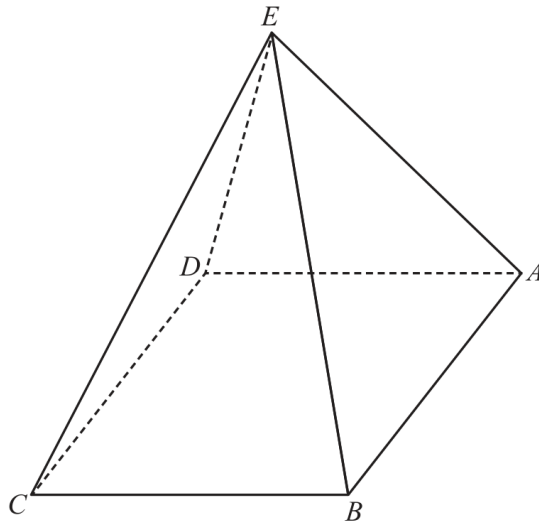
Use 3D trigonometry to obtain a terminal volume

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 26

4MA1-2H Q24

UNIT 1
4 marks**24** The diagram shows a square-based pyramid $ABCDE$ Diagram **NOT**
accurately drawn

$$EA = EB = EC = ED$$

 M is the centre of the horizontal square base $ABCD$ Q is the midpoint of AB Angle $EQM = 80^\circ$

$$EA : AB = n : 1$$

Find the value of n

Give your answer correct to 3 significant figures.

 $n = \dots\dots\dots$ **(Total for Question 24 is 4 marks)**

WORKING SPACE

Question 17

4MA1-1H Q21

MODULAR UNIT 1

5 marks

- 21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

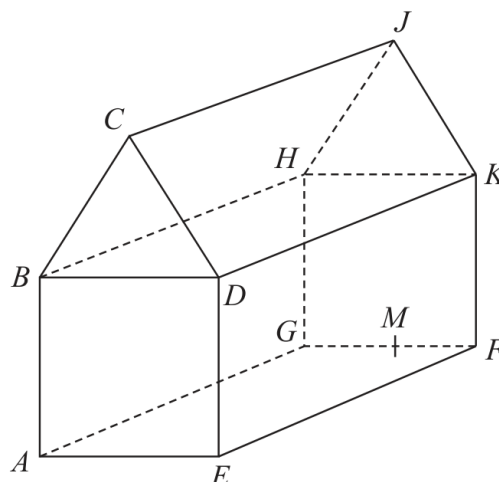


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

$$\text{Angle } MAJ = y^\circ$$

$$\text{Given that } \tan y^\circ = T$$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T = \dots\dots\dots$

Working space for Question 17

Continue your method

UNIT 1**5 marks**

WORKING SPACE

19 The diagram shows a cuboid $ABCDEFGH$.

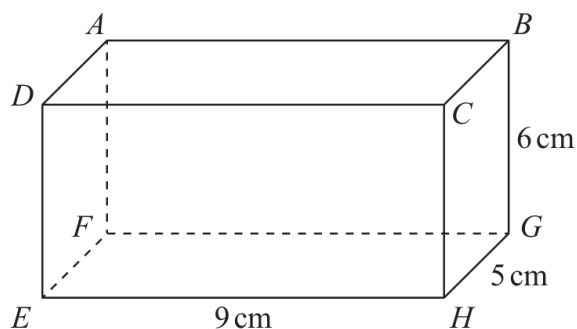


Diagram **NOT**
accurately drawn

$EH = 9\text{ cm}$, $HG = 5\text{ cm}$ and $GB = 6\text{ cm}$.

Work out the size of the angle between AH and the plane $EFGH$.
Give your answer correct to 3 significant figures.

o

YOUR WORKING

- (b) Calculate the size of the angle between AF and the plane $ADEH$.
Give your answer correct to one decimal place.

.....
(3)

YOUR WORKING

22 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

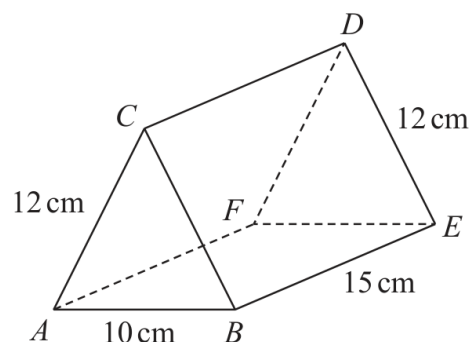


Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

YOUR WORKING

18 The diagram shows cuboid $ABCDEFGH$.

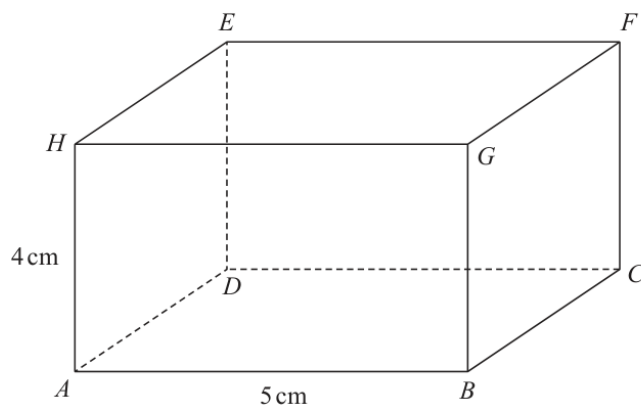


Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm}$$

$$AH = 4 \text{ cm}$$

The size of the angle between CH and the plane $ABCD$ is 35°

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

.....cm³

(Total for Question 18 is 5 marks)

YOUR WORKING

23 The diagram shows a solid prism $ABCDEFGHJ$

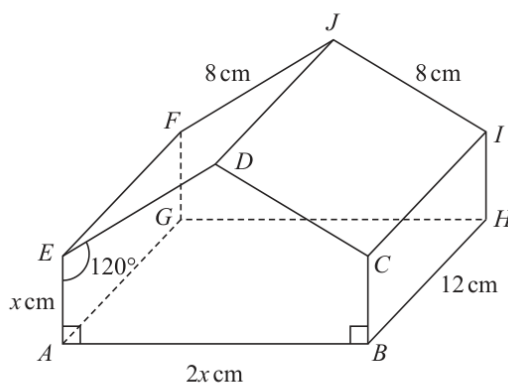


Diagram **NOT**
accurately drawn

The prism is such that each cross section is a pentagon where

$$AE = BC = x \text{ cm} \quad AB = 2x \text{ cm} \quad ED = CD = 8 \text{ cm}$$

$$\text{angle } EAB = \text{angle } CBA = 90^\circ \quad \text{angle } AED = \text{angle } BCD = 120^\circ$$

Given that $AG = BH = EF = DJ = CI = 12 \text{ cm}$

calculate the angle that AJ makes with the base $ABHG$ of the prism.

Give your answer correct to 3 significant figures.



YOUR WORKING

(Total for Question 23 is 5 marks)

YOUR WORKING

$$AB = 10 \text{ cm} \qquad BG = 14 \text{ cm}$$
$$BC = \frac{1}{5}AD$$

Calculate the volume of the prism.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... cm³

(Total for Question 26 is 5 marks)

YOUR WORKING
